

PROTOTYPE OVERVIEW — PREPARED FOR REVIEW

TD One — Workshop

A clickable, working prototype that puts one mining tyre or rim repair job on one screen — from the mine site to dispatch — built for Tyre Doctor, an Australian mining tyre and rim repair business. This document explains what the problem was, what was asked for, what this prototype actually does, and how to walk someone through it.

What's in this document

1. The problem, in Tyre Doctor's own words

2. What they actually asked for

3. What this prototype is — and isn't

4. What each screen shows

5. How to walk through it, step by step

6. Questions to be ready for

7. Known gaps — said plainly

Read this before a meeting about TD One and you should be able to explain, in plain words, what it is, why it exists, and how to click through it yourself — without needing anyone else in the room to fill in gaps.

SECTION 1

The problem, in Tyre Doctor's own words

Tyre Doctor is a second-generation, family-owned Australian business. They repair the giant tyres used on mining haul trucks, loaders and dozers — a single tyre weighs about **six tonnes** and costs enough that mines pay specialists to repair it rather than replace it. They also crack-test steel wheel rims, which is safety-critical work: a cracked rim under pressure can fail explosively. They run this out of **eight workshop branches** across Australia.

Today, the information for one repair job is scattered across four different places:

1. A row in a spreadsheet (Smartsheet)
2. Photos in a WhatsApp thread, found again later by quote number
3. A paper sheet, handwritten on the workshop floor and photographed
4. The quote itself, sitting in their accounting/ERP system, NetSuite

"At the moment it's going into four different things and we're pulling it together at the end."

— Tyre Doctor, describing how a job's information is currently tracked

What that actually causes, day to day:

- The mine calls sales asking **"where's my tyre?"** — sales then has to call the workshop to find out, because nobody has one place to look.
- At the end of a job, someone has to manually rebuild the whole story of the repair from four different sources, to write a report.
- The operations superintendent builds a branch-performance spreadsheet by hand, in his own time, because there's no system that already knows this.

"Out of them six tyres, there's five we can repair."

— Tyre Doctor, describing how one site visit produces a whole batch of separate jobs at once

That one line matters more than it looks: **one visit to a mine site is one pile of tyres, which becomes many separate repair jobs, but still just one quote and one pickup.** Whatever gets built has to keep that batch-to-individual-job relationship straight, or the numbers won't match what Tyre Doctor actually does.

SECTION 2

What they actually asked for

Tyre Doctor's own description of the goal was simple: they want something like a **pizza tracker** — the kind of screen that shows exactly which stage an order is at, in plain language, without anyone having to phone and ask. Applied here, that means: one tyre or rim, one file, visible the whole way from the mine site to dispatch — to the workshop, to sales, and to the mine itself.

The hardest requirement they gave, in their own words:

"You're not dealing with the most tech-savvy guys, so it's got to be simple and easy to use, otherwise they just put it aside and go back to a piece of paper."

That one sentence is the real design brief. It's not a request for more features — it's a warning that if this isn't obviously faster and easier than a phone photo and a paper sheet, the workshop floor will quietly stop using it and go back to what they know. Every screen in this prototype was built with that warning in mind: big touch targets, one clear action per screen, plain English instead of jargon, and nothing that adds a step without removing a bigger one.

What was explicitly out of scope

Tyre Doctor was just as clear about what this should *not* do:

- **No prices anywhere.** Quoting stays in NetSuite. This product only ever shows a quote *number* and a status.
- **No claim of being "connected" or "synced" to anything.** If a field looks like it came from NetSuite, it's because a person typed it in — not because two systems are talking to each other.
- **No AI, anywhere** — no auto-detection, no "AI found this", no confidence scores.
- **No on-site fleet inspection tool.** That's the fitters' daily tread-and-pressure rounds, and it's explicitly a later, separate piece of work.
- **No login.** A simple "who are you today?" role picker stands in for it in this prototype.

SECTION 3

What this prototype is — and isn't

It IS: a clickable, working front-end that shows exactly how the finished product would behave — real interactions, a real state machine, a real spreadsheet-reading feature, real printable certificates — all running in a web browser with no setup.

It is NOT: connected to Tyre Doctor's actual NetSuite, Smartsheet, or email systems. There is no server and no database — it's a front end only, seeded with realistic made-up data so it can be clicked through convincingly. Building the real backend, and connecting it to Tyre Doctor's existing systems, is the next phase of work, not something this prototype claims to already do.

The two repair "rails"

Everything in TD One follows one of two paths, because Tyre Doctor actually does two different services:

TYRE REPAIR	RIM TESTING (NDT)
Triaged at the mine → collected → received → washed → inspected → cut out → repaired → cooked (hot-cured) → quoted → dispatched.	Notified → received → blasted → crack-tested → clean rims are blasted/tested/painted; cracked rims are repaired and re-tested → certified (two certificates) → dispatched.

The single most important behaviour in the whole product is the **escalation**: if, once a tyre is actually inspected or cut open, the damage turns out worse than what was quoted, that one tyre **pauses** — sales is told immediately — while every other tyre in the same batch keeps moving. Nothing waits on the whole batch just because one tyre needs a second look.

SECTION 4

What each screen shows

In the order you'd naturally click through them:

ANYONE

1. Landing page

The front door. States the problem in one sentence and has one button: "Open the workshop." A second link, "See the customer view", shows what a mine sees — useful to show the contrast early.

Look for: how little it asks of a first-time visitor before they're inside the product.

ANYONE

2. Who are you today? ROLE PICKER

Stands in for a login. Instead of a username and password, a person just taps who they are today — workshop supervisor, repairer, NDT technician, sales, and so on — and lands on the screen that role actually uses.

Look for: this is a deliberate simplification for the demo, not a claim that the real product will never need proper accounts.

WORKSHOP FLOOR

3. Branch Console THE MAIN SCREEN

The actual "pizza tracker." Everything currently on the floor, laid out as lanes matching where it physically is — just arrived, being washed, being inspected, in the oven, ready to go — plus a matching set of lanes for rims being crack-tested. A "Needs a person" band at the top surfaces anything a human has to make a decision on. Search finds any job instantly by quote number or serial.

Look for: click any card to open its full history; nothing here shows a price.

REPAIR MANAGER, AT THE MINE

4–6. Field capture TYRE

The phone screen used out at the mine, before a tyre ever reaches a workshop: visit details, then a fast repeating loop (three photos per tyre, serial, repairable yes/no), then a visit summary that's sent to sales. Built to survive a bad mobile signal — work queues quietly and sends itself, it never tells the fitter they're "offline."

Look for: this is the screen the "must be simple" warning applies to most directly.

WORKSHOP FLOOR

7–9. Job File, stage actions, and escalation

One tyre's entire life on one page: every stage it has passed through, who did it, when, and the photo that proves it — plus the cook record and what was actually measured versus what was originally quoted. "Advance this tyre" moves it to its next stage and only asks for what that exact stage needs. "Flag worse than quoted" is the escalation described above — it opens a real email preview showing exactly what sales would receive.

Look for: the required fields differ per stage — the product won't let a stage be skipped without a supervisor's logged reason.

SUPERVISOR / NDT TECHNICIAN

10–12. Rim intake, NDT bay, certificates RIM

Rim intake actually reads a real spreadsheet file the mine sends over — no shortcuts, it genuinely parses the rows, sorting each rim into "ready to test" or "needs a person to look at it first" (duplicate entry, missing data, odd serial, etc.), each with a plain-English reason. Every rim then goes through crack testing and, on a pass, gets two numbered, watermarked certificates.

Look for: the certificates carry a visible "PLACEHOLDER TEMPLATE" watermark — Tyre Doctor's real certificate design hasn't been supplied yet, so this is honestly marked as a stand-in.

SUPERINTENDENT / BRANCH MANAGERS

13–14. Job report and Branch Performance

A printable, one-page report of a single job's whole story. Separately, a cross-branch view: repair mix per branch, hours spent, and how long jobs sit at each stage — the same numbers the superintendent currently builds by hand, generated automatically instead.

THE MINE (CUSTOMER)

15. Customer Tracker EXTERNAL, NO LOGIN

What a mine sees if given a link: plain milestones only — received, being prepared, cooking, boxed up, dispatched — with a few photos and certificates once issued. No price, no internal detail, and nothing that hasn't actually happened yet.

BONUS, ADDED AFTER THE INITIAL BUILD

16. Process flow diagram

A separate, editable diagram (reachable from a "Process flow" link in the header on most screens) that lays out this exact process left to right, matching the real state machine the app runs on — not a separate guess at it. It opens collapsed to a high-level view; click any step to open its detail, or use "Expand all detail" / "Show high level" to toggle everything at once. It can be edited directly (add, rename, delete, reconnect steps, add sticky notes) and downloaded as an image or a re-loadable file — useful for talking through the process in a meeting without touching the running prototype.

SECTION 5

How to walk through it, step by step

A suggested order for demonstrating the whole thing in one sitting, about 10–15 minutes:

1 Open the workshop

Start at Landing, click "Open the workshop", then pick **Workshop supervisor** on the Who-are-you screen. You land on the Branch Console for Mackay, the busiest branch.

2 Show the floor, not a form

Point out the lanes (on the way → wash → inspect → oven → ready) and the "Needs a person" band. Click "Show me around" to run the guided tour if you want the room to see it narrated. Tap "Mark received" on a tyre to show a live status change with an undo option.

3 Open one job's whole story

Click any tyre card to open its Job File. Walk through the story spine (each stage, timestamped, with its photo), the cook record, and "what we measured." Click "Advance this tyre" to show a stage actually being completed, including a required photo or measurement.

4 Show the escalation — this is the important one

Open a tyre that's already flagged (or flag one live), and click through to "See what sales gets told." This is the single behaviour Tyre Doctor cares most about: one tyre pauses, everything else keeps moving, and sales gets a clear notification with no phone call needed.

5 Go to the mine site

Switch to Field (phone-width screen). Do a quick capture: three photos, serial, repairable yes/no, save, and show the "waiting for signal" queue banner — the message that reassures a fitter their work is safe even with no bars.

6 Bring in a real spreadsheet

Go to Rim intake and actually drop in the sample rim-returns file. Show the real 18-rows-in, 13-jobs-created, 5-held split, and open one held row to show the plain-English reason. Walk one rim through NDT to its two certificates (point out the watermark).

7 Zoom out to the branch and the mine's own view

Open Branch Performance to show the cross-branch numbers, then open the Customer Tracker link to show exactly what a mine would see — and how much less detail it deliberately shows than the workshop view.

8 Reset and hand over the keyboard

Tap "Reset demo" (twice, to confirm) to put everything back to its starting state, then let the room click through it themselves if there's time.

SECTION 6

Questions to be ready for

Q. Is this connected to our real NetSuite / Smartsheet / email?

A. No. This is a front-end prototype with made-up but realistic data, so it can be clicked through convincingly. Every quote number, photo and job in it is invented. Wiring it to Tyre Doctor's actual systems is a separate, later phase.

Q. Does it show prices?

A. No, deliberately. Quoting stays in NetSuite; this product only ever holds a quote number and a status (Initial / Revised / Final).

Q. Is there any AI in this?

A. No. Nothing is auto-detected or auto-classified. Every field is either typed by a person or calculated from rules Tyre Doctor gave us (like the damage-size thresholds), and every one of those rules is written down and shown with its source.

Q. Does this replace the fitters' daily tyre inspection rounds?

A. No — that's explicitly out of scope for this piece of work. This only covers a tyre or rim once it's been flagged for repair or testing.

Q. Why isn't there a proper login?

A. For this prototype stage, a simple role picker does the same demonstration job as a login without the overhead of building real accounts before the workflow itself is agreed. A real login is a normal, expected part of building the finished product.

Q. Are the certificates the real Tyre Doctor design?

A. No — we haven't been given Tyre Doctor's real certificate template yet, so every certificate in this prototype is clearly watermarked "PLACEHOLDER TEMPLATE" rather than presenting a guess as the real thing.

Q. How long would the real, connected version take to build?

A. Not something this document estimates — that depends on scope decisions (which systems it connects to, hosting, who manages it) that haven't been made yet. This prototype exists to make those decisions concrete, not to replace them.

SECTION 7

Known gaps — said plainly

In the spirit of not overselling this: a short, honest list of what's simplified or not yet finished in this pass, so nobody is surprised by it live.

- **No backend.** Refreshing the browser resets nothing mid-session, but there's no database — this is a front end only, by design, at this stage.
- **Certificates are a placeholder design**, watermarked as such, until Tyre Doctor's real template is available.
- **Keyboard-only navigation between cards on the floor view isn't built yet** — the floor screens are designed mouse/tap-first, matching how they'll actually be used on the shop floor.
- **The guided tour on the rim-intake screen covers 2 of the 3 steps originally planned** — the third lived on a screen that was split out into its own page during the build.

TD One (Workshop) — prototype overview. Prepared to help someone outside this project understand what it is, why it exists, and how to talk about it confidently in a meeting.